MULTIVARIABLE CALCULUS • UNIT 5.01

C Triple Integrals, Cylindrical & Spherical Coordinates

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 C	Q2 C	Q3 D	Q4 D	Q5 C
Q6 A	Q7 C	Q8 B	Q9 D	Q10 C
Q11 B	Q12 A	Q13 A	Q14 C	Q15 A
Q16 C	Q17 B	Q18 C	Q19 B	Q20 D
Q21 A	Q22 C	Q23 C	Q24 A	Q25 A

Triple Integrals, Cylindrical & Spherical Coordinates Practice Exercises

1 MCQ

Evaluate $\iiint_B xyz\,dV$ over the box $B = [0,3] \times [0,1] \times [0,3]$.

A $\frac{81}{4}$

B $\frac{81}{16}$

C $\frac{81}{8}$ ✓

D $\frac{153}{8}$

WORKED SOLUTION & VERIFICATION

Iterate: $\int_0^3 \int_0^1 \int_0^3 xyz\,dx\,dy\,dz = \frac{81}{8}$.

2 MCQ

Evaluate $\iiint_B x+y+z\,dV$ over the box $B = [0,3] \times [0,2] \times [0,1]$.

A 36

B 9

C 18 ✓

D 24

WORKED SOLUTION & VERIFICATION

Iterate: $\int_0^1 \int_0^2 \int_0^3 (x+y+z)\,dx\,dy\,dz = 18$.

3 MCQ

Evaluate $\iiint_B xyz\,dV$ over the box $B = [0,2] \times [0,1] \times [0,2]$.

A 4

B 1

C 6

D 2 ✓

WORKED SOLUTION & VERIFICATION

Iterate: $\int_0^2 \int_0^1 \int_0^2 xyz\,dx\,dy\,dz = 2$.

4 MCQ

Evaluate $\iiint_B yz^2\,dV$ over the box $B = [0,1] \times [0,1] \times [0,3]$.

- **A** 9
- **B** $\frac{9}{4}$
- **C** $\frac{15}{2}$
- **D** $\frac{9}{2}$ ✓

WORKED SOLUTION & VERIFICATION

Iterate: $\int_0^3 \int_0^1 \int_0^1 yz^2\,dx\,dy\,dz = \frac{9}{2}$.

5 MCQ

Evaluate $\iiint_B x^2 z\,dV$ over the box $B = [0,1] \times [0,1] \times [0,3]$.

- **A** 3
- **B** $\frac{3}{4}$
- **C** $\frac{3}{2}$ ✓
- **D** $\frac{9}{2}$

WORKED SOLUTION & VERIFICATION

Iterate: $\int_0^3 \int_0^1 \int_0^1 x^2 z\,dx\,dy\,dz = \frac{3}{2}$.

6 MCQ

Evaluate $\iiint_B xy + z\,dV$ over the box $B = [0,1] \times [0,1] \times [0,3]$.

- **A** $\frac{21}{4}$ ✓
- **B** $\frac{21}{2}$
- **C** $\frac{21}{8}$
- **D** $\frac{33}{4}$

WORKED SOLUTION & VERIFICATION

Iterate: $\int_0^3 \int_0^1 \int_0^1 (xy + z)\,dx\,dy\,dz = \frac{21}{4}$.

7 MCQ

Evaluate $\iiint_B xy + z\,dV$ over the box $B = [0,1] \times [0,3] \times [0,3]$.

- **A** $\frac{81}{2}$
- **B** $\frac{81}{8}$
- **C** $\frac{81}{4}$ ✓
- **D** $\frac{117}{4}$

WORKED SOLUTION & VERIFICATION

Iterate: $\int_0^3 \int_0^3 \int_0^1 (xy + z)\,dx\,dy\,dz = \frac{81}{4}$.

8 MCQ

Evaluate $\iiint_B x + y + z\, dV$ over the box $B = [0,3] \times [0,3] \times [0,1]$.

- A 63
- B $\frac{63}{2}$ ✓
- C $\frac{63}{4}$
- D $\frac{81}{2}$

WORKED SOLUTION & VERIFICATION

Iterate: $\int_0^1 \int_0^3 \int_0^3 (x + y + z)\, dx\, dy\, dz = \frac{63}{2}$.

9 MCQ

Consider the vector field $\mathbf{u}(s, y, z) = \langle 22s,\ 7y,\ 7z \rangle$ on R^3. Compute the scalar divergence $\nabla \cdot \mathbf{u}$.

- A 40
- B 29
- C 58
- D **36** ✓

WORKED SOLUTION & VERIFICATION

The divergence is $22 + 7 + 7 = 36$.

10 MCQ

Consider the solid region E bounded below by the cone $z = \sqrt{x^2 + y^2}$ and above by the sphere $x^2 + y^2 + z^2 = 16$. The volume of E is given by the triple integral $V = \iiint_E dV$. If E is expressed in spherical coordinates, the integral becomes $\int_\alpha^\beta \int_\gamma^\delta \int_{\varrho_1}^{\varrho_2} \varrho^2 \sin\phi\, d\varrho\, d\phi\, d\theta$. Determine the value of α, the lower limit of integration for θ.

- A $\frac{\pi}{2}$
- B π
- C **0** ✓
- D 2π

WORKED SOLUTION & VERIFICATION

The region E is a spherical cone. The upper bound is a sphere of radius 4 centered at the origin, so ϱ ranges from 0 to 4. The lower bound is the cone $z = \sqrt{x^2 + y^2}$. In spherical coordinates, $z = \varrho \cos\phi$ and $\sqrt{x^2 + y^2} = \varrho \sin\phi$. Thus, the cone equation becomes $\varrho \cos\phi = \varrho \sin\phi$. For $\varrho \neq 0$, this simplifies to $\cos\phi = \sin\phi$, which means $\tan\phi = 1$. In the range $0 \le \phi \le \pi$, this occurs at $\phi = \frac{\pi}{4}$. Therefore, ϕ ranges from 0 to $\frac{\pi}{4}$. Since the solid region is symmetric about the z-axis and covers the entire 'spiky' part of the cone and sphere, θ ranges from 0 to 2π. Thus, $\alpha = 0$, $\beta = 2\pi$, $\gamma = 0$, $\delta = \frac{\pi}{4}$, and $\varrho_1 = 0$, $\varrho_2 = 4$. The lower limit of integration for θ is $\alpha = 0$.

11 MCQ

A solid region E is defined by the inequalities $x^2 + y^2 \leq 16$, $0 \leq z \leq 5$. The volume of this region can be computed using a triple integral. In cylindrical coordinates, the region E is described by $0 \leq r \leq R$, $0 \leq \theta \leq 2\pi$, and $0 \leq z \leq H$. What is the value of R?

A. 2
B. 4 ✓
C. 8
D. 16

WORKED SOLUTION & VERIFICATION

The given region is a cylinder with radius R_{cyl} and height H_{cyl}. The inequality $x^2 + y^2 \leq 16$ describes a circle of radius 4 centered at the origin in the xy-plane. When converting to cylindrical coordinates, $x^2 + y^2$ becomes r^2. Thus, the inequality $r^2 \leq 16$ implies $r \leq 4$. Therefore, in cylindrical coordinates, the radial component of the region is described by $0 \leq r \leq 4$. The question states that the region is described by $0 \leq r \leq R$, so $R = 4$. The inequality $0 \leq z \leq 5$ defines the height of the cylinder, which is $H = 5$. The limits for θ are $0 \leq \theta \leq 2\pi$ as the region is a full circle.

12 MCQ

Consider the solid region E in the first octant bounded by the coordinate planes and the surface $z = 4 - x^2 - y^2$. The volume of this region is given by the triple integral $V = \iiint_E dV$. We are given that the volume can also be expressed in cylindrical coordinates as $V = \int_0^{\pi/2} \int_0^{\sqrt{a}} \int_0^{4-r^2} r dz dr d\theta$. Find the value of the parameter a.

A. 4 ✓
B. 2
C. 16
D. 8

WORKED SOLUTION & VERIFICATION

The solid region E is in the first octant and is bounded by $z = 0$, $x = 0$, $y = 0$ and $z = 4 - x^2 - y^2$. In cylindrical coordinates, $x^2 + y^2 = r^2$, so the surface is $z = 4 - r^2$. Since the region is in the first octant, $0 \leq \theta \leq \pi/2$. The lower bound for z is $z = 0$. The upper bound for z is $4 - r^2$.

To determine the bounds for r, we look at the projection of the solid onto the xy-plane. This projection is determined by the intersection of $z = 4 - x^2 - y^2$ with the xy-plane (i.e., $z = 0$). This gives $0 = 4 - x^2 - y^2$, or $x^2 + y^2 = 4$. In the first quadrant of the xy-plane, this is a quarter circle of radius 2. Thus, $0 \leq r \leq 2$.

The given integral in cylindrical coordinates is $V = \int_0^{\pi/2} \int_0^{\sqrt{a}} \int_0^{4-r^2} r dz dr d\theta$. Comparing the limits for r, we have $\sqrt{a} = 2$. Squaring both sides, we get $a = 4$.

13 MCQ

Let the multivariable function u be defined by $u(s, y) = 14s^2y + 24y^2$. Compute the partial derivative $\frac{\partial u}{\partial s}(6, 10)$.

A. 1680 ✓
B. 1656
C. 1684
D. 1694

WORKED SOLUTION & VERIFICATION

Differentiating with respect to s gives $28sy$, which equals 1680 at $(6, 10)$.

14 MCQ

Use spherical coordinates to find the volume of the upper half of the ball $x^2 + y^2 + z^2 \le 4$.

A 4π

B $\frac{32\pi}{3}$

C $\frac{16\pi}{3}$ ✓

D $\frac{8\pi}{3}$

WORKED SOLUTION & VERIFICATION

$dV = \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$; integrating over the region gives $\frac{1}{2} \cdot \frac{4}{3}\pi(2)^3 = \frac{16\pi}{3}$.

15 MCQ

Use spherical coordinates to find the volume of the first-octant part of the ball $x^2 + y^2 + z^2 \le 16$.

A $\frac{32\pi}{3}$ ✓

B $\frac{16\pi}{3}$

C $\frac{256\pi}{3}$

D $\frac{64\pi}{3}$

WORKED SOLUTION & VERIFICATION

$dV = \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$; integrating over the region gives $\frac{1}{8} \cdot \frac{4}{3}\pi(4)^3 = \frac{32\pi}{3}$.

16 MCQ

Use cylindrical coordinates to find the volume of the cone $\frac{2}{4}\sqrt{x^2 + y^2} \le z \le 2$.

A $\frac{8\pi}{3}$

B 32π

C $\frac{32\pi}{3}$ ✓

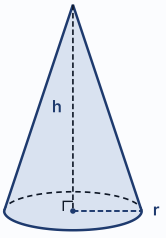
D 16π

WORKED SOLUTION & VERIFICATION

$V = \int_0^{2\pi} \int_0^4 \int_{\frac{1}{2}r}^2 r \, dz \, dr \, d\theta = \frac{32\pi}{3}$.

17 MCQ

Use cylindrical coordinates to find the volume of the cone $\frac{1}{3}\sqrt{x^2 + y^2} \le z \le 1$.



A 9π

B 3π ✓

C $\frac{9\pi}{2}$

D π

WORKED SOLUTION & VERIFICATION

$V = \int_0^{2\pi} \int_0^3 \int_{\frac{1}{3}r}^1 r \, dz \, dr \, d\theta = 3\pi$.

18 MCQ

Use cylindrical coordinates to find the volume of the cone $\frac{4}{4}\sqrt{x^2+y^2} \le z \le 4$.

A $\frac{16\pi}{3}$

B 64π

C $\frac{64\pi}{3}$ ✓

D 32π

WORKED SOLUTION & VERIFICATION

$V = \int_0^{2\pi}\int_0^4\int_{1r}^4 r\,dz\,dr\,d\theta = \frac{64\pi}{3}$.

19 MCQ

Use spherical coordinates to find the volume of the upper half of the ball $x^2+y^2+z^2 \le 9$.

A $\frac{27\pi}{2}$

B 18π ✓

C 36π

D 9π

WORKED SOLUTION & VERIFICATION

$dV = \varrho^2 \sin\phi\, d\varrho\, d\phi\, d\theta$; integrating over the region gives $\frac{1}{2}\cdot\frac{4}{3}\pi(3)^3 = 18\pi$.

20 MCQ

Use spherical coordinates to find the volume of the upper half of the ball $x^2+y^2+z^2 \le 16$.

A $\frac{256\pi}{3}$

B $\frac{64\pi}{3}$

C 32π

D $\frac{128\pi}{3}$ ✓

WORKED SOLUTION & VERIFICATION

$dV = \varrho^2 \sin\phi\, d\varrho\, d\phi\, d\theta$; integrating over the region gives $\frac{1}{2}\cdot\frac{4}{3}\pi(4)^3 = \frac{128\pi}{3}$.

21 MCQ

Use cylindrical coordinates to find the volume of the cone $\frac{3}{4}\sqrt{x^2+y^2} \le z \le 3$.

A 16π ✓

B 48π

C 24π

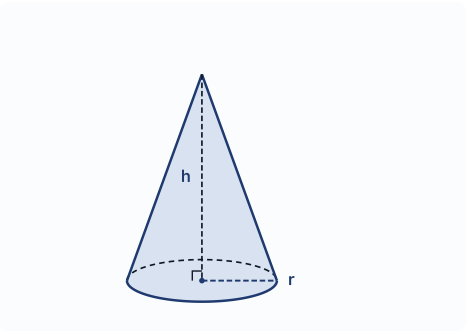
D 4π

WORKED SOLUTION & VERIFICATION

$V = \int_0^{2\pi}\int_0^4\int_{\frac{3}{4}r}^3 r\,dz\,dr\,d\theta = 16\pi$.

22 MCQ

Use cylindrical coordinates to find the volume of the cone $\frac{4}{1}\sqrt{x^2+y^2} \le z \le 4$.



A 4π

B 2π

C $\frac{4\pi}{3}$ ✓

D $\frac{8\pi}{3}$

WORKED SOLUTION & VERIFICATION

$V = \int_0^{2\pi}\int_0^1\int_{4r}^4 r\,dz\,dr\,d\theta = \frac{4\pi}{3}$.

23 MCQ

Use cylindrical coordinates to find the volume of the cone $\frac{2}{3}\sqrt{x^2+y^2} \le z \le 2$.

A 18π

B 9π

C 6π ✓

D 2π

WORKED SOLUTION & VERIFICATION

$V = \int_0^{2\pi}\int_0^3\int_{\frac{2}{3}r}^2 r\,dz\,dr\,d\theta = 6\pi$.

24 MCQ

Use cylindrical coordinates to find the volume of the cone $\frac{4}{2}\sqrt{x^2+y^2} \le z \le 4$.

A $\frac{16\pi}{3}$ ✓

B $\frac{8\pi}{3}$

C 16π

D 8π

WORKED SOLUTION & VERIFICATION

$V = \int_0^{2\pi}\int_0^2\int_{2r}^4 r\,dz\,dr\,d\theta = \frac{16\pi}{3}$.

25 MCQ

Use cylindrical coordinates to find the volume of the cone $\frac{3}{2}\sqrt{x^2+y^2} \le z \le 3$.

A 4π ✓

B 12π

C 6π

D 2π

WORKED SOLUTION & VERIFICATION

$V = \int_0^{2\pi}\int_0^2\int_{\frac{3}{2}r}^3 r\,dz\,dr\,d\theta = 4\pi$.